Lecture 2

Linear Regression

Table of contents

1

Linear Regression

Linear Regression

Linear regression is a statistical technique used to model the relationship between a dependent variable (also called the target or response variable) and one or more independent variables (also called predictors or features)

Linear Regression

For example, you might want to predict a person's height (in inches) from his weight (in pounds).

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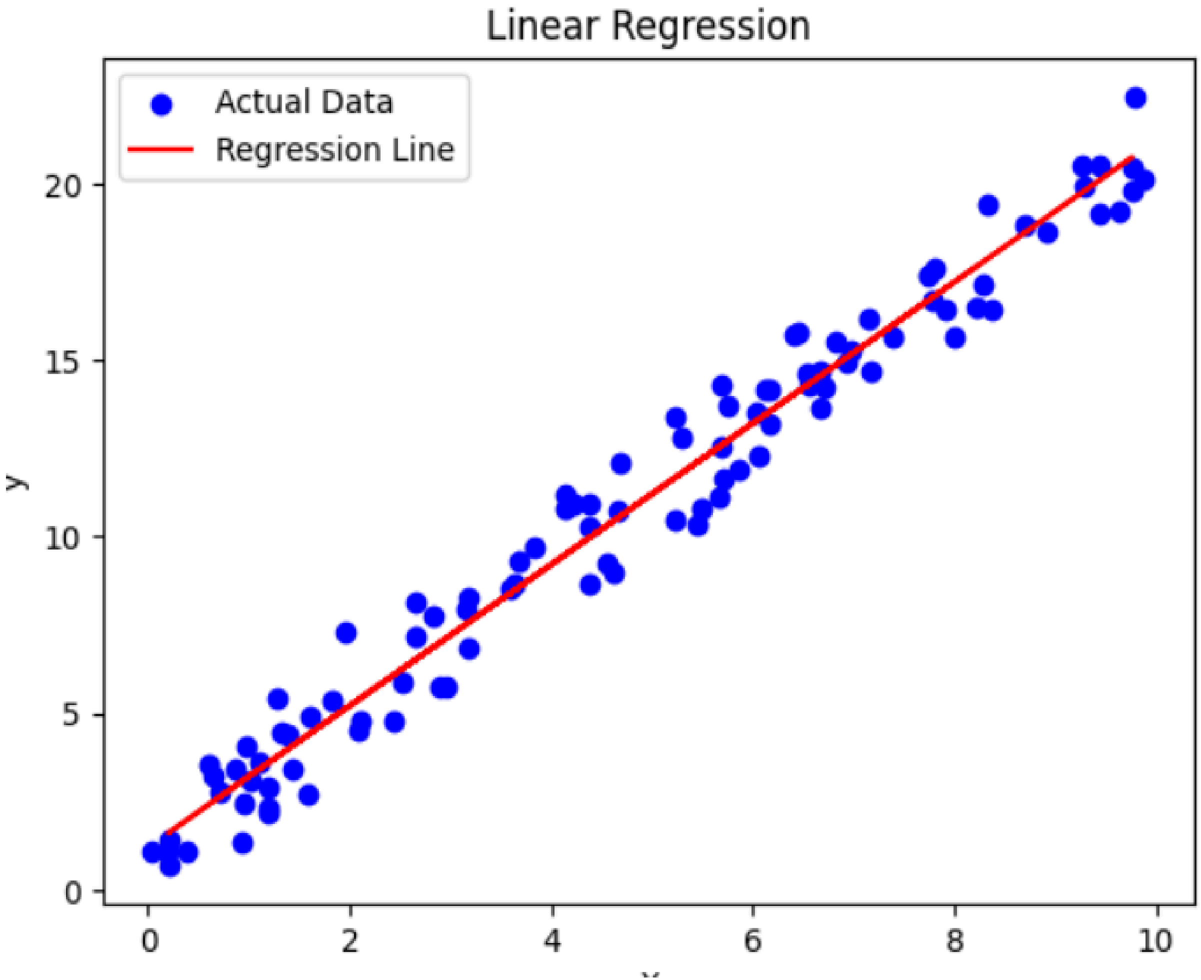
Linear Regression

$$y=\beta_0+\beta_1x+\epsilon$$

$$Y = mX + C$$



Linear Regression

$$\beta_1 = \frac{\sum(x_i - \bar{x})(y_i - \bar{y})}{\sum(x_i - \bar{x})^2}$$

$y=\beta_0+\beta_1 x+\epsilon$

$Y = mX + C$

$$\beta_0 = \bar{y} - \beta_1 \bar{x}$$

Linear Regression

Suppose You have to Build a linear regression model to predict the pizza prices from Following data

Pizza Size (x)	Pizza Price(y)	X-	y-	(X-)*(y-	$(X-)^2$
8	10	-2	-3	6	4
10	13	0	0	0	0
12	16	2	3	6	4

$$\bar{X} = \frac{8+10+12}{3} = 10 \qquad \bar{y} = \frac{10+13+16}{3} = 13$$

Linear Regression

Pizza Size (x)	Pizza Price(y)	X-	y-	(X-)*(y-	(X-)^2
8	10	-2	-3	6	4
10	13	0	0	0	0
12	16	2	3	6	4

$$\beta_1 = \frac{\sum(x_i - \bar{x})(y_i - \bar{y})}{\sum(x_i - \bar{x})^2} = \frac{6+6}{4+4} = \frac{12}{8} = 1.5$$

$$\beta_0 = \bar{y} - \beta_1 \bar{x}$$

Linear Regression

$$Y = 1.5X - 2$$

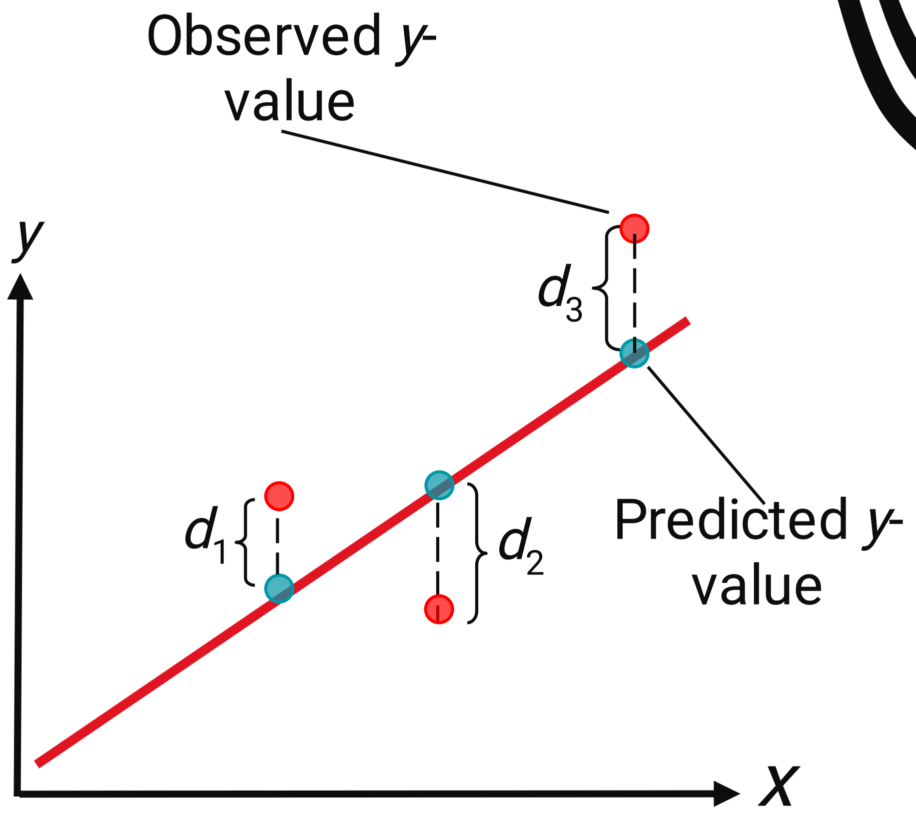
Linear Regression

Real = Y

Predicted = $\hat{Y}$

$$\text{Error } d_1 = Y_i - \hat{Y}$$

$$\text{Total Error} = \sum(Y_i - \hat{Y})$$



Linear Regression

$Y = \beta_0 + \beta_1 x$

$$\text{Total Error} = \sum(Y_i - (\beta_0 + \beta_1 x))$$

Sum of Square Error

$$S(B_0, B_1) = \sum(Y_i - (\beta_0 + \beta_1 x))^2$$

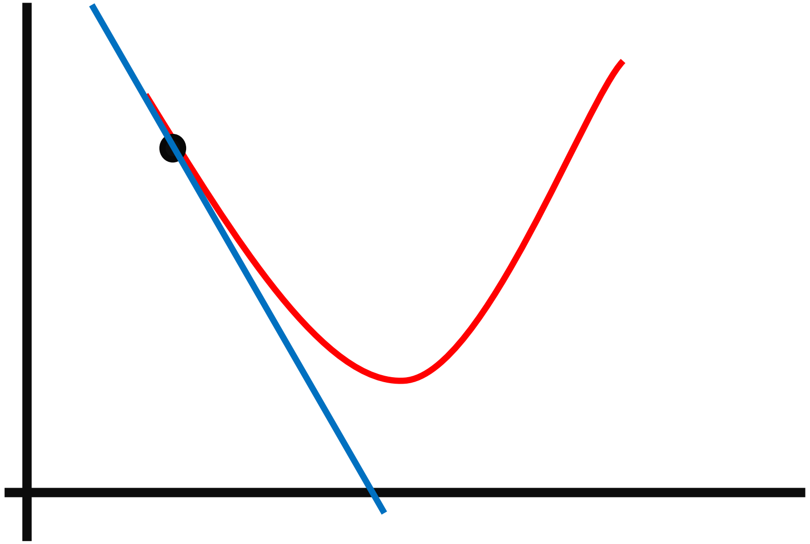
(i) Partial Derivative with Respect to β_0

$$\frac{\partial S(\beta_0, \beta_1)}{\partial \beta_0} = \frac{\partial}{\partial \beta_0} \sum_{i=1}^{n} (y_i - (\beta_0 + \beta_1 x_i))^2$$

Linear Regression

By applying the chain rule:

$$\frac{\partial S(\beta_0, \beta_1)}{\partial \beta_0} = -2\sum_{i=1}^{n}(y_i - (\beta_0 + \beta_1 x_i))$$

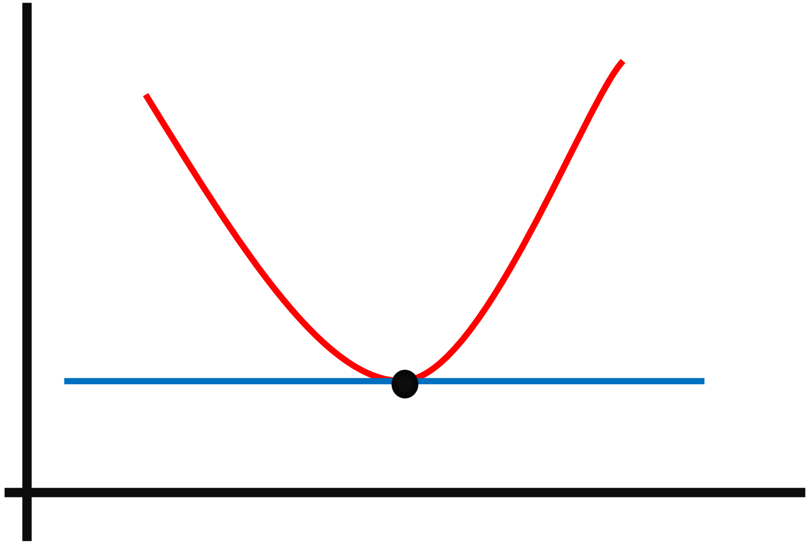


Linear Regression

Setting the derivative equal to zero:

$$\sum_{i=1}^{n} (y_i - \beta_0 - \beta_1 x_i) = 0$$

$$\sum_{i=1}^{n} y_i = n\beta_0 + \beta_1 \sum_{i=1}^{n} x_i$$



Linear Regression

From this, we can express β_0 as:

$$\beta_0 = \frac{\sum_{i=1}^{n} y_i - \beta_1 \sum_{i=1}^{n} x_i}{n}$$

Or equivalently:

$$\beta_0 = \bar{y} - \beta_1 \bar{x}$$

Linear Regression

(ii) Partial Derivative with Respect to β_1

Now we differentiate with respect to β_1:

$$\frac{dE}{dm} = \frac{\partial}{\partial m}\sum (y_i - (\beta_0 + m\, x_i))^2$$

$$\frac{dE}{dm} = \sum 2(Y_i - mx_i \; - \bar{y} + m\bar{X})(-x_i + \dot{x}) = 0$$

$$\sum (Y_i - mx_i \; - \bar{y} + m\bar{X})(x_i - \dot{x}) = 0$$

$$\sum [(Y_i - \bar{y}) - m(xi \; - \bar{X})](x_i - \dot{x}) = 0$$

Linear Regression

$$\sum[(Y_i - \bar{y}) - m(xi \ - \bar{X})](xi - \dot{x}) = 0$$

$$\sum[(Y_i - \bar{y})(xi - \dot{x}) - m(x_i \ - \bar{X})^2] \ = 0$$

$$\sum[(Y_i - \bar{y})(xi - \dot{x}) - \sum m(x_i \ - \bar{X})^2] \ = 0$$

$$\sum[(Y_i - \bar{y})(xi - \bar{x}) = \sum m(x_i \ - \bar{X})^2$$

$$\sum[(Y_i - \bar{y})(xi - \bar{x}) = m \ \sum(x_i \ - \bar{X})^2$$

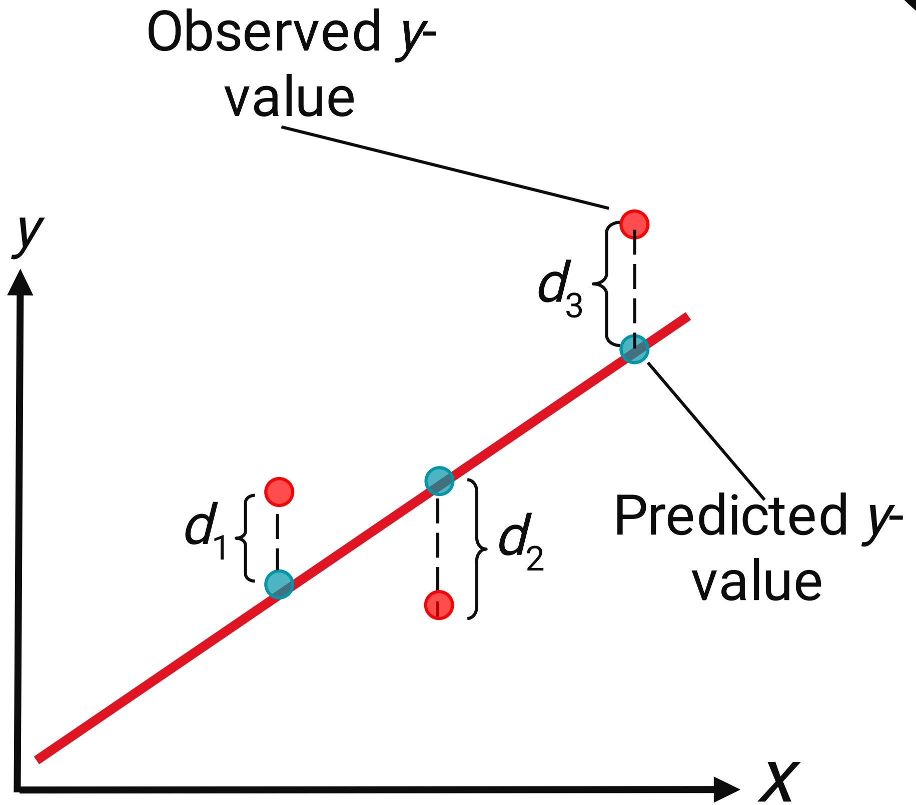
$$m = \frac{\sum[(Y_i - \bar{y})(xi - \bar{x})}{\sum(x_i \ - \bar{X})^2}$$

Gradient Descent

$$y = wx_i + b$$

$$\text{Total Error} = \sum (Y_i - \hat{Y})$$

$$\text{Loss Function } l = \frac{1}{n} \sum (Y_i - (wx_i + b))^2$$



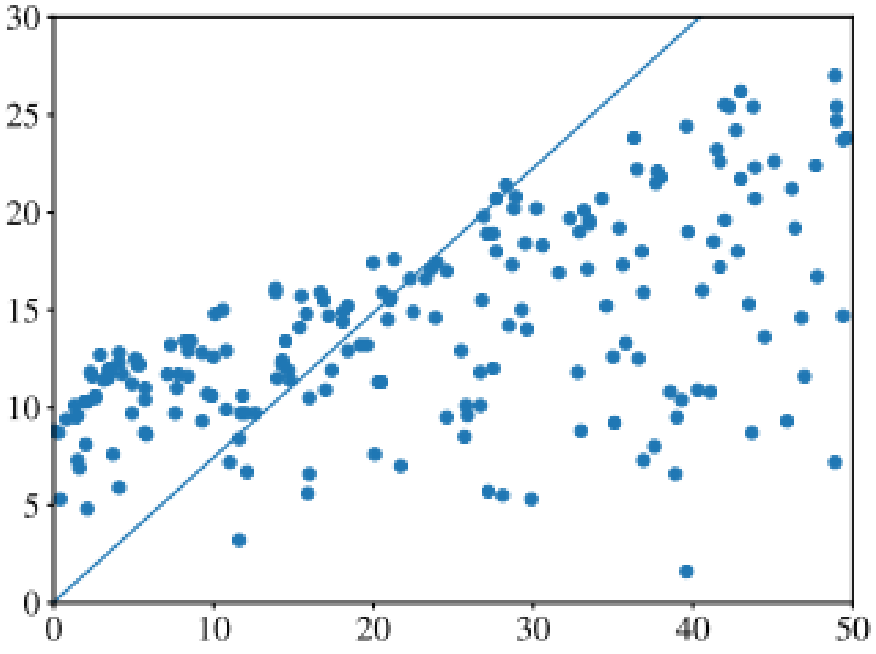
Gradient Descent

w ← 0 and b ← 0

$$w \leftarrow w - \alpha \frac{\partial l}{\partial w};$$

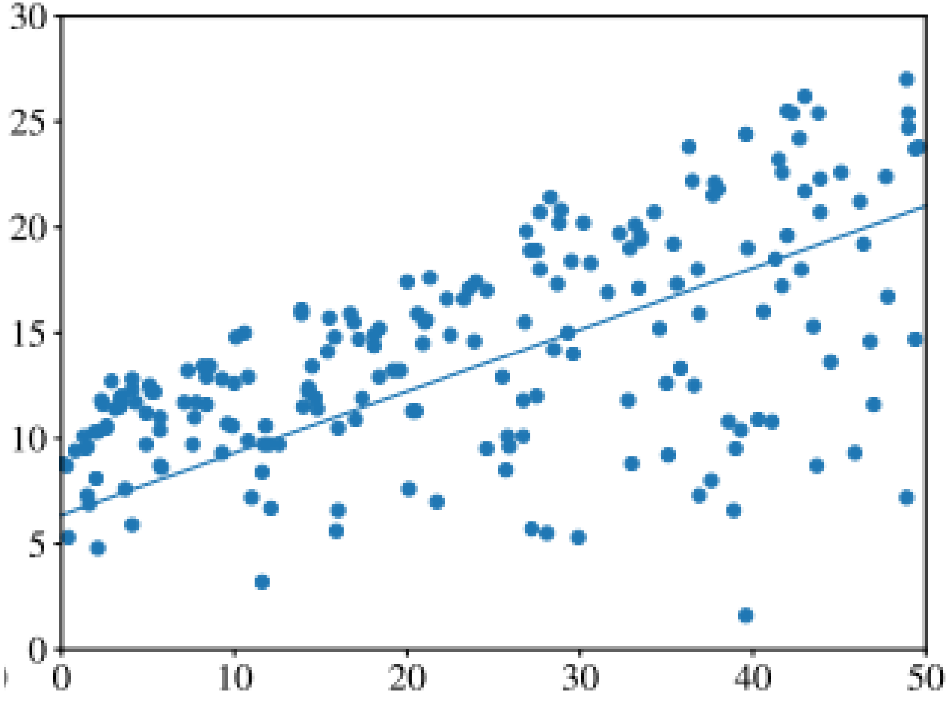
$$b \leftarrow b - \alpha \frac{\partial l}{\partial b}.$$

Gradient Descent



Epoch 0

Gradient Descent



Epoch 3000

Gradient Descent

The cost function $J(m, c)$ for linear regression is the **mean squared error (MSE)**:

$$J(m,c) = \frac{1}{2n} \sum_{i=1}^{n} (Y_i - (mX_i + c))^2$$

$$m := m - \alpha \frac{\partial}{\partial m} J(m,c)$$

$$c := c - \alpha \frac{\partial}{\partial c} J(m,c)$$

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Gradient Descent

X	Y	Y(Predicted)
1	1	0
2	2	0
3	3	0
4	4	0

$Y = mx + c$
Let,

$m = 0$
$C = 0$

Gradient Descent

Where α is the **learning rate**, and the partial derivatives of the cost function are:

$$\frac{\partial}{\partial m} J(m,c) = -\frac{1}{n}\sum_{i=1}^{n} X_i \left(Y_i - (mX_i + c)\right)$$

$$\frac{\partial}{\partial c} J(m,c) = -\frac{1}{n}\sum_{i=1}^{n} \left(Y_i - (mX_i + c)\right)$$

$$\frac{\partial}{\partial m} J(m,c) = -\frac{1}{n}\sum X_i \left(Y_i - \hat{Y}_i\right)$$

Gradient Descent

$$\frac{\partial}{\partial m} J(m,c) = -\frac{1}{n} \sum X_i \left(Y_i - \hat{Y}_i \right)$$

$$= -\frac{1}{4}(1(1-0) + 2(2-0) + 3(3-0) + 4(4-0)) = -\frac{1}{4}(1+4+9+16) = -7.5$$

Gradient Descent

$$\frac{\partial}{\partial c}J(m,c) = -\frac{1}{n}\sum\left(Y_i - \hat{Y}_i\right)$$

$$= -\frac{1}{4}\left((1-0)+(2-0)+(3-0)+(4-0)\right) = -\frac{1}{4}(1+2+3+4) = -2.5$$

Gradient Descent

$$m_1 = m_0 - \alpha \times \frac{\partial}{\partial m} J(m, c)$$

$$m_1 = 0 - 0.01 \times (-7.5) = 0 + 0.075 = 0.075$$

$$c_1 = c_0 - \alpha \times \frac{\partial}{\partial c} J(m, c)$$

$$c_1 = 0 - 0.01 \times (-2.5) = 0 + 0.025 = 0.025$$

Gradient Descent

Iteration 2:

1. **Compute New Predictions:**

$$\hat{Y}_i = 0.075 \times X_i + 0.025$$

- For $X_1 = 1$: $\hat{Y}_1 = 0.075 \times 1 + 0.025 = 0.1$
- For $X_2 = 2$: $\hat{Y}_2 = 0.075 \times 2 + 0.025 = 0.175$
- For $X_3 = 3$: $\hat{Y}_3 = 0.075 \times 3 + 0.025 = 0.25$
- For $X_4 = 4$: $\hat{Y}_4 = 0.075 \times 4 + 0.025 = 0.325$

Gradient Descent

2. **Compute Gradients:**

$$\frac{\partial}{\partial m}J(m,c) = -\frac{1}{4}(1(1-0.1)+2(2-0.175)+3(3-0.25)+4(4-0.325))$$

$$= -\frac{1}{4}(1(0.9)+2(1.825)+3(2.75)+4(3.675)) = -\frac{1}{4}(0.9+3.65+8.25+14.7) = -6.875$$

$$\frac{\partial}{\partial c}J(m,c) = -\frac{1}{4}((1-0.1)+(2-0.175)+(3-0.25)+(4-0.325))$$

$$= -\frac{1}{4}(0.9+1.825+2.75+3.675) = -2.2875$$

3. **Update Parameters:**

$$m_2 = m_1 - \alpha \times \frac{\partial}{\partial m}J(m,c)$$

$$m_2 = 0.075 - 0.01 \times (-6.875) = 0.075 + 0.06875 = 0.14375$$

$$c_2 = c_1 - \alpha \times \frac{\partial}{\partial c}J(m,c)$$

$$c_2 = 0.025 - 0.01 \times (-2.2875) = 0.025 + 0.022875 = 0.047875$$

Gradient Descent

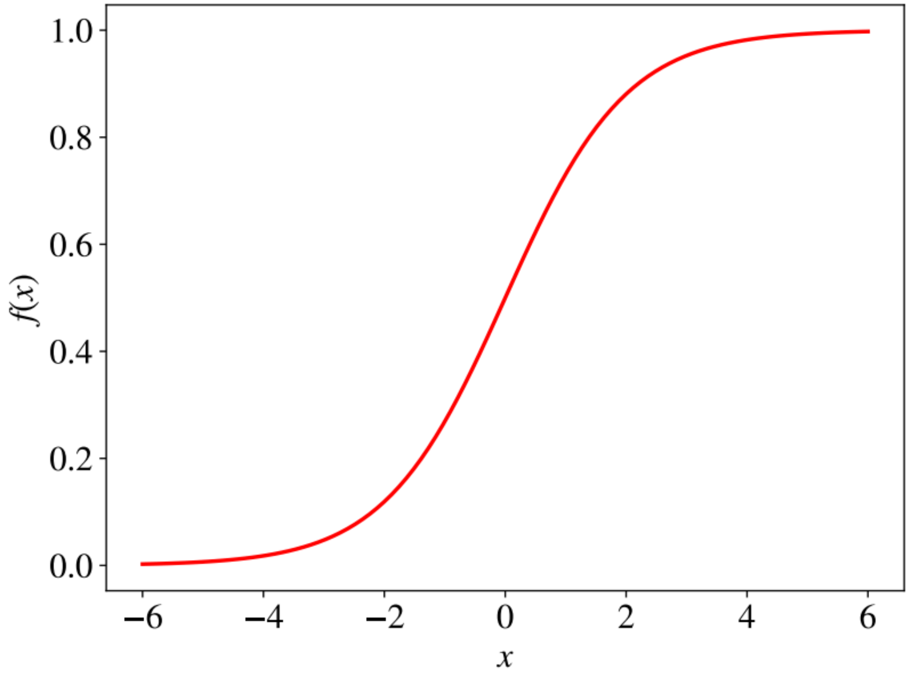
Result After 2 Iterations:

- New values: $m = 0.14375$, $c = 0.047875$
- The line after 2 iterations is: $Y = 0.14375X + 0.047875$

Logistic Regression

$$f(x) = \frac{1}{1+e^{-x}}$$

$$f(x) = \frac{1}{1+e^{-(wx+b)}}$$



Logistic Regression

$$f(x) = \frac{1}{1+e^{-(wx+b)}}$$

$$h_\theta(x) = \frac{1}{1 + e^{-(\theta_0 + \theta_1 x)}}$$

Logistic Regression

$$h_\theta(x) = \frac{1}{1 + e^{-(\theta_0 + \theta_1 x)}}$$

Step 2: Binary Cross-Entropy Cost Function

The cost function for logistic regression, also known as binary cross-entropy, is:

$$J(\theta) = -\frac{1}{n}\sum_{i=1}^{n}\left(y^{(i)}\log(h_\theta(x^{(i)})) + (1 - y^{(i)})\log(1 - h_\theta(x^{(i)}))\right)$$

Logistic Regression

$$h_\theta(x) = \frac{1}{1 + e^{-(\theta_0 + \theta_1 x)}}$$

Step 2: Binary Cross-Entropy Cost Function

The cost function for logistic regression, also known as binary cross-entropy, is:

$$J(\theta) = -\frac{1}{n}\sum_{i=1}^{n}\left(y^{(i)}\log(h_\theta(x^{(i)})) + (1 - y^{(i)})\log(1 - h_\theta(x^{(i)}))\right)$$

Step 3: Gradient Descent

We will use gradient descent to minimize the cost function and update the parameters θ_0 and θ_1.

The gradient update rules for logistic regression are:

$$\theta_j := \theta_j - \alpha \frac{\partial J(\theta)}{\partial \theta_j}$$

Where:

- α is the learning rate.
- $\frac{\partial J(\theta)}{\partial \theta_j}$ is the gradient of the cost function with respect to θ_j.

For logistic regression, the gradients are computed as:

$$\frac{\partial J(\theta)}{\partial \theta_0} = \frac{1}{n} \sum_{i=1}^{n} \left(h_\theta(x^{(i)}) - y^{(i)} \right)$$

$$\frac{\partial J(\theta)}{\partial \theta_1} = \frac{1}{n} \sum_{i=1}^{n} \left(\left(h_\theta(x^{(i)}) - y^{(i)} \right) x^{(i)} \right)$$

Logistic Regression

Hour Studied	Pass/ Fail	Y(Predicted)
2	0	.5
3	0	.5
5	1	.5
7	1	.5

Logistic Regression

$$h_\theta(x) = \frac{1}{1 + e^{-(\theta_0 + \theta_1 x)}}$$

1. **Compute Hypothesis for Each Example:**

$$h_\theta(2) = \frac{1}{1 + e^{-(0+0\times 2)}} = \frac{1}{1+1} = 0.5$$

$$h_\theta(3) = \frac{1}{1 + e^{-(0+0\times 3)}} = \frac{1}{1+1} = 0.5$$

$$h_\theta(5) = \frac{1}{1 + e^{-(0+0\times 5)}} = \frac{1}{1+1} = 0.5$$

$$h_\theta(7) = \frac{1}{1 + e^{-(0+0\times 7)}} = \frac{1}{1+1} = 0.5$$

Logistic Regression

$$J(\theta) = -\frac{1}{n}\sum_{i=1}^{n}\left(y^{(i)}\log(h_\theta(x^{(i)})) + (1-y^{(i)})\log(1-h_\theta(x^{(i)}))\right)$$

The cost for all 4 examples is:

$$J(\theta) = -\frac{1}{4}\left[0\log(0.5) + (1-0)\log(1-0.5) + 0\log(0.5) + (1-0)\log(1-0.5)\right]$$

$$= -\frac{1}{4}\left[\log(0.5) + \log(0.5) + \log(0.5) + \log(0.5)\right]$$

$$= -\frac{1}{4} \times 4 \times (-0.693) = 0.693$$

Logistic Regression

$$\frac{\partial J(\theta)}{\partial \theta_0} = \frac{1}{n}\sum_{i=1}^{n}\left(h_\theta(x^{(i)}) - y^{(i)}\right)$$

$$\frac{\partial J(\theta)}{\partial \theta_1} = \frac{1}{n}\sum_{i=1}^{n}\left(\left(h_\theta(x^{(i)}) - y^{(i)}\right)x^{(i)}\right)$$

3. **Compute the Gradients:**

The gradients for θ_0 and θ_1 are:

$$\frac{\partial J(\theta)}{\partial \theta_0} = \frac{1}{4}\left[(0.5-0)+(0.5-0)+(0.5-1)+(0.5-1)\right] = \frac{1}{4}\times(-1) = -0.25$$

$$\frac{\partial J(\theta)}{\partial \theta_1} = \frac{1}{4}\left[(0.5-0)\times 2+(0.5-0)\times 3+(0.5-1)\times 5+(0.5-1)\times 7\right]$$

$$= \frac{1}{4}\left[1+1.5-2.5-3.5\right] = \frac{1}{4}\times(-3.5) = -0.875$$

Logistic Regression

4. **Update Parameters:**

Using the learning rate $\alpha = 0.1$:

$$\theta_0 := \theta_0 - 0.1 \times (-0.25) = 0 + 0.025 = 0.025$$

$$\theta_1 := \theta_1 - 0.1 \times (-0.875) = 0 + 0.0875 = 0.0875$$

Thank You